

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	Buvanesh Y
Register Number	192372367

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

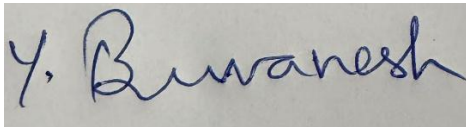
Total Marks: 25

Code of Conduct

I **Buvanesh Y (192372367)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.
4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

Thought process

Thought process

Cloud Application-Based Questions

1. Auto-Scaling & Load Balancing for E-Commerce

The Problem:

An e-commerce company like Amazon or Flipkart sees normal traffic on regular days but faces 10x-50x spikes during Black Friday, Diwali Sales, or flash deals. Pre-provisioning servers for peak load wastes money 350 days a year. Under-provisioning causes crashes exactly when revenue is highest.

How Auto-Scaling Works:

Cloud Auto Scaling Groups continuously monitor metrics like CPU usage, request count, and response time. When traffic rises above a threshold (say CPU > 70%), the system automatically launches new server instances within 2-3 minutes. When traffic drops, extra instances are terminated. You only pay for what you use. Scaling policies can be:

- **Reactive:** Triggers when a metric crosses a threshold
- **Scheduled:** Pre-scales before known events (e.g., midnight sale launch)
- **Predictive:** Uses ML to forecast traffic and scale ahead of time

How Load Balancing Works:

An Application Load Balancer (ALB) sits in front of all servers and distributes incoming requests evenly. If one server becomes slow or crashes, the load balancer stops sending traffic to it and redirects requests to healthy servers — users never see the failure. It also handles SSL termination, reducing processing load on individual servers.

Reducing Downtime & Improving User Experience:

Before cloud scaling, a traffic spike would overwhelm servers, causing slow page loads and checkout failures — costing thousands in lost sales per minute. With auto-scaling and load balancing, new capacity appears automatically before servers get overwhelmed. Users experience consistent page load times even during peak demand. Checkout flows remain smooth, cart abandonment stays low, and revenue is protected.

Role of Monitoring Tools:

Amazon CloudWatch, Datadog, or New Relic continuously track server health, response times, and error rates. Dashboards give real-time visibility. Alerts notify engineers before problems affect users. Monitoring feeds data directly into auto-scaling decisions — without monitoring, the system cannot know when to scale. During a Flipkart Big Billion Day sale, CloudWatch triggers scale-out at the first sign of rising CPU, ensuring capacity is ready before users notice any slowdown.

2. Cloud Backup Storage — Durability, Security & Cost

Durability & Data Redundancy:

Cloud storage services like Amazon S3 are designed for 99.999999999% (11 nines) durability. Every file uploaded is automatically replicated across a minimum of three Availability Zones (separate physical data centers within a region). If one data center experiences a fire or power failure, your data remains intact in the other two. Cross-region replication can additionally copy data to a completely different geography — protecting against regional disasters like floods or earthquakes. Versioning preserves every previous copy of a file, so accidental overwrites or deletions are fully recoverable.

Encryption & Access Control:

All backup data should be encrypted at rest using AES-256 and in transit using TLS 1.3. AWS S3 supports Server-Side Encryption automatically — no extra configuration needed from developers. For sensitive data, Customer Managed Keys (CMK) through AWS KMS give full cryptographic control, meaning even cloud provider employees cannot read your data. Access Control is enforced through IAM policies — only specific roles (e.g., the backup service account) can write to the backup bucket, and only the disaster recovery team can read from it. S3 Object Lock prevents any user from deleting backups during a defined retention period, protecting against both accidents and ransomware.

Cost Advantages Over Traditional Storage:

On-premises backup infrastructure requires purchasing tape drives or NAS servers, maintaining a secondary disaster recovery site, hiring staff to manage hardware, and refreshing equipment every 3-5 years. Cloud storage eliminates all of this. You pay only for what you store — typically \$0.023 per GB per month for S3 Standard. S3 Intelligent Tiering automatically moves infrequently accessed backups to cheaper storage classes, cutting costs by up to 60-70% without any manual management. No upfront capital expense — the cost is fully operational.

Disaster Recovery Example:

A hospital's patient database is accidentally corrupted during a software update. Because continuous backups were enabled with point-in-time recovery, the database is restored to its exact state from 10 minutes before the corruption — losing minimal data. Route 53 failover switches application traffic to the restored database in a secondary region. Total recovery time: under 30 minutes. Without cloud backup, this could have meant days of data loss and regulatory penalties.

3. CDN & Global Deployment for Mobile Apps

How CDNs Improve Performance:

A Content Delivery Network is a globally distributed network of edge servers (Points of Presence) placed in cities worldwide — Mumbai, Singapore, Frankfurt, New York, São Paulo. When a user requests content, they connect to the nearest edge server rather than a central origin server thousands of miles away. This reduces latency from 200-300ms (cross-continent) to under 20ms (local edge). Static assets — images, CSS, JavaScript, video segments — are cached at edge servers and served instantly without hitting the origin at all.

How Edge Servers Enhance User Experience:

Edge servers do more than just cache files. They handle SSL termination locally, reducing connection setup time. They compress responses on the fly. They perform intelligent routing — if the nearest edge server is overloaded, the CDN automatically routes the request to the next closest one. For a gaming application, edge servers handle matchmaking and game state for players in their region, ensuring low ping times. For a streaming app, video chunks are pre-fetched and cached at the edge so playback is smooth even on slower connections.

Cloud Provider Global Availability:

AWS CloudFront has 400+ edge locations globally. Azure CDN and Google Cloud CDN offer similar coverage. These integrate directly with cloud services — content in S3 is instantly distributable through CloudFront with a few clicks. Cloud providers also offer anycast routing, meaning a single IP address automatically routes traffic to the geographically closest data center.

Real-World Example — Streaming & Gaming:

Consider a live streaming platform like Hotstar during an IPL cricket match with 25 million concurrent viewers across India. Without CDN, all 25 million requests would hammer origin servers simultaneously — causing instant overload. With CDN, the live video stream is segmented into small chunks (HLS format) and cached at dozens of Indian edge locations (Mumbai, Chennai, Delhi, Bangalore). Each city's viewers get content from their nearest edge node. Origin servers handle only the initial stream source — the CDN multiplies that capacity across millions of users. The result: smooth 1080p streaming with under 3 seconds of delay for every viewer, regardless of their location.

4. Cloud Security & Compliance for Financial Institutions

Regulatory Compliance Support:

Financial institutions must comply with regulations like PCI-DSS (payment card security), SOX (financial reporting), GDPR (data privacy), and RBI guidelines. Cloud providers like AWS, Azure, and GCP invest billions in compliance certifications and maintain pre-built compliance frameworks. AWS provides AWS Artifact — a portal with compliance reports and certifications available for download to submit during audits. Services like AWS Config continuously check whether your cloud resources comply with defined rules and flag violations automatically. This reduces audit preparation from weeks to hours.

Logging & Monitoring for Security:

Every action in the cloud — who accessed what data, which API was called, what configuration changed — is recorded by services like AWS CloudTrail and Azure Monitor. These logs are immutable, tamper-proof, and stored separately from production systems. A SIEM platform (Splunk or Microsoft Sentinel) aggregates all logs and detects suspicious patterns — for example, a user downloading 50,000 customer records at 2am triggers an immediate alert. Amazon GuardDuty uses machine learning to detect unusual behavior like impossible logins (same account logging in from India and US within minutes) or cryptocurrency mining on compromised servers.

Encryption for Financial Data:

All transaction data encrypted at rest (AES-256) and in transit (TLS 1.3). Tokenization replaces sensitive card numbers with random tokens in databases — even if the database is breached, attackers get useless

tokens. Hardware Security Modules (HSMs) — available as AWS CloudHSM — manage encryption keys with FIPS 140-2 Level 3 certification, the highest standard for financial-grade key management.

Data Sovereignty:

Regulatory requirements may prohibit customer data from leaving a specific country. Cloud providers solve this through region locking — you deploy resources only in approved regions (e.g., AWS Mumbai region for Indian RBI compliance) and enforce Service Control Policies that literally prevent data from being transferred to other regions. Data residency policies are documented and auditable.

Real-World Example:

When a customer makes a credit card payment, the transaction data is encrypted end-to-end. The payment processor running on AWS uses CloudHSM to sign the transaction, CloudTrail logs every step, GuardDuty monitors for fraud patterns, and the entire transaction — from swipe to approval — happens in under 2 seconds while maintaining complete audit trails for regulatory review.

5. Containers, Kubernetes & Microservices

Containers vs Virtual Machines:

A Virtual Machine (VM) includes a full operating system (several GBs), virtualized hardware, and then your application on top. This takes minutes to start and consumes significant memory even when idle. A container shares the host machine's operating system kernel — it packages only the application and its dependencies. This makes containers extremely lightweight (megabytes vs gigabytes), start in seconds, and allows dozens of containers to run on the same server where only a few VMs would fit. A single EC2 instance that runs 5 VMs can run 50-100 containers, dramatically improving resource utilization and reducing infrastructure cost.

How Kubernetes Simplifies Deployment & Scaling:

Kubernetes (K8s) is a container orchestration platform that automates the heavy lifting of running containers at scale. You declare the desired state — "I want 10 instances of the payment service running" — and Kubernetes ensures that state is always maintained. If a container crashes, Kubernetes restarts it automatically. If a node (server) fails, Kubernetes reschedules its containers on healthy nodes — zero manual intervention. Deployments support rolling updates — new versions are rolled out gradually, and if errors are detected, Kubernetes automatically rolls back to the previous version. Horizontal Pod Autoscaler adds more containers when CPU rises and removes them when load drops, just like VM auto-scaling but faster (seconds vs minutes).

Portability Across Cloud Environments:

Because containers package the application along with all its dependencies and configuration, they run identically on a developer's laptop, an on-premises server, AWS, Azure, or GCP. This eliminates environment-specific bugs and enables true multi-cloud flexibility — if AWS pricing increases or a region goes down, workloads can migrate to Azure without code changes. Helm charts package Kubernetes deployment configurations as reusable templates, making deployment to any cluster a single command.

Microservices Example — Food Delivery App:

Consider an app like Zomato broken into independent microservices — User Service, Restaurant Service,

Order Service, Payment Service, Delivery Tracking Service, and Notification Service. Each runs as its own set of containers in Kubernetes. During lunchtime, only the Order and Payment services see traffic spikes — Kubernetes scales just those services up while others remain at minimal capacity, saving cost. The Notification Service can be updated (new push notification format) and redeployed without touching or restarting any other service. If the Delivery Tracking Service crashes due to a bug, the rest of the app continues working — users can still browse restaurants and place orders. This isolation, independent scaling, and independent deployment is the core advantage of combining microservices with containerization.